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Student Name: Bipashna Thapa

London Met ID:23050399

College ID: NP01CP4A230137

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I confirm that I understand my coursework needs to be submitted online via MST Classroom under the relevant module page before the deadline in order for my assignment to be accepted and marked. I am fully aware that late submissions will be treated as non-submissions and a mark of zero will be awarded.

Table of Contents

1. Textual Analysis *4*

2. Assumptions *8*

3. Normalization *8*

UNF 8

1NF 9

2NF 9

3NF 10

Table of Figures

Figure 1 User and Booking 4

Figure 2 Booking and Show 4

Figure 3 Show and Movie..... 5

Figure 4 Show and Hall 5

Figure 5 Theatre and Hall..... 6

Figure 6 Theatre and Movie 6

Figure 7 User and Movie 7

Figure 8 Movie and Show Time 7

Figure 9 Initial ERD 11

1. Textual Analysis

1. User and Booking

A user can make **one or multiple bookings**, and each booking belongs to **a single user**.

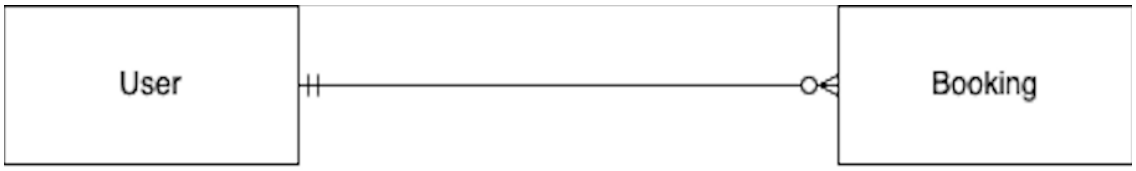

Figure 1 User and Booking

2. Booking and Show

A booking is made for **one show**, and a show can have **one or multiple bookings**.

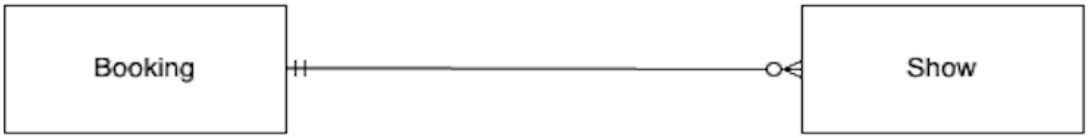

Figure 2 Booking and Show

2. **Show and Movie**

A show is associated with **one movie**, and a movie can have **one or multiple shows**.

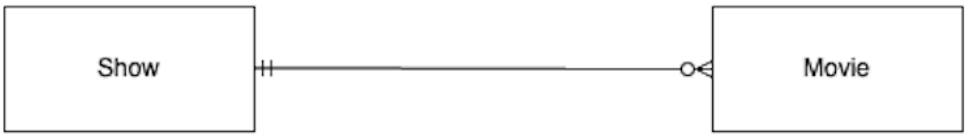

Figure 3 Show and Movie

4. Show and Hall

A show takes place in **one hall**, and a hall can host **one or multiple shows**.

Figure 4 Show and Hall

5. Hall and Theatre

A theatre can have **one or multiple halls**, and each hall belongs to **a single theatre**.

Figure 5 Theatre and Hall

6. Theatre and Movie

A theatre can screen **one or multiple movies**, and a movie can be screened in **one or multiple theatres**.

Figure 6 Theatre and Movie

7. User and Movie

A user can book tickets for **one or multiple movies**, and a movie can be booked by **one or multiple users**.

Figure 7 User and Movie

8. Movie and Show Time

A movie can have **multiple show times**, but each show time is associated with **a single movie**.

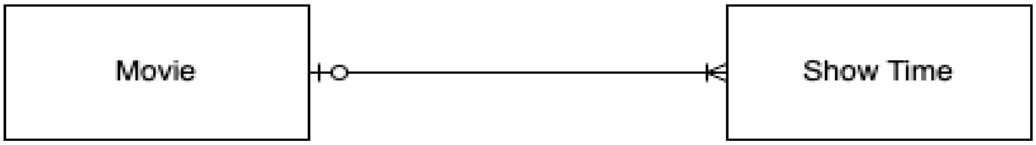

Figure 8 Movie and Show Time

2. Assumptions

The assumptions for the system are given below:

1. One user can book **multiple tickets for the same movie**.
2. One user can also book tickets for **different movies**.
3. A theatre can have **multiple halls**.
4. Each hall can host **only one show at a time**.

3. Normalization

UNF

MOVIE_TICKETING_UNF(<u>userID,</u> username, address, {movieID, movieTitle, duration, language, genre, releaseDate, theatreID, theatreName, theatreCity, {hallID, hallNumber, hallCapacity, showID, showTime, showDate, ticketPrice}})

In the above UNF process, all attributes with repeating groups are included in a single relation.

1NF

USER_1NF -> (userID, username, address)

MOVIE_1NF -> (movieID, movieTitle, duration, language, genre, releaseDate)

THEATRE_1NF -> (theatreID, theatreName, theatreCity)

HALL_1NF -> (hallID, hallNumber, hallCapacity, theatreID*)

SHOW_1NF -> (showID, hallID*, movieID*, showTime, showDate, ticketPrice)

BOOKING_1NF-> (userID*, showID*)

2NF

USER_2NF -> (userID, username, address)

MOVIE_2NF -> (movieID, movieTitle, duration, language, genre, releaseDate)

THEATRE_2NF -> (theatreID, theatreName, theatreCity)

HALL_THEATRE_2NF -> (hallID*, theatreID*)

SHOW_2NF -> (showID, showTime, showDate, ticketPrice)

SHOW_HALL_2NF -> (showID*, hallID*)

SHOW_MOVIE_2NF-> (showID*, movieID*)

3NF

USER_3NF -> (<u>userID</u>, username, address)

MOVIE_3NF -> (<u>movieID</u>, movieTitle, duration, language, genre, releaseDate)

THEATRE -> (<u>theatreID,</u> theatreName)

THEATRE_LOCATION-> (<u>theatreID</u>, theatreCity)

HALL -> (<u>hallID,</u> hallNumber, hallCapacity)

HALL_THEATRE_3NF -> (<u>hallID*, theatreID*</u>)

SHOW_HALL_3NF->(<u>showID*, hallID*)</u>

SHOW_MOVIE_3NF-> (<u>showID*, movieID*</u>)

BOOKING -> (<u>userID*, showID*</u>)

4. **Initial ERD**

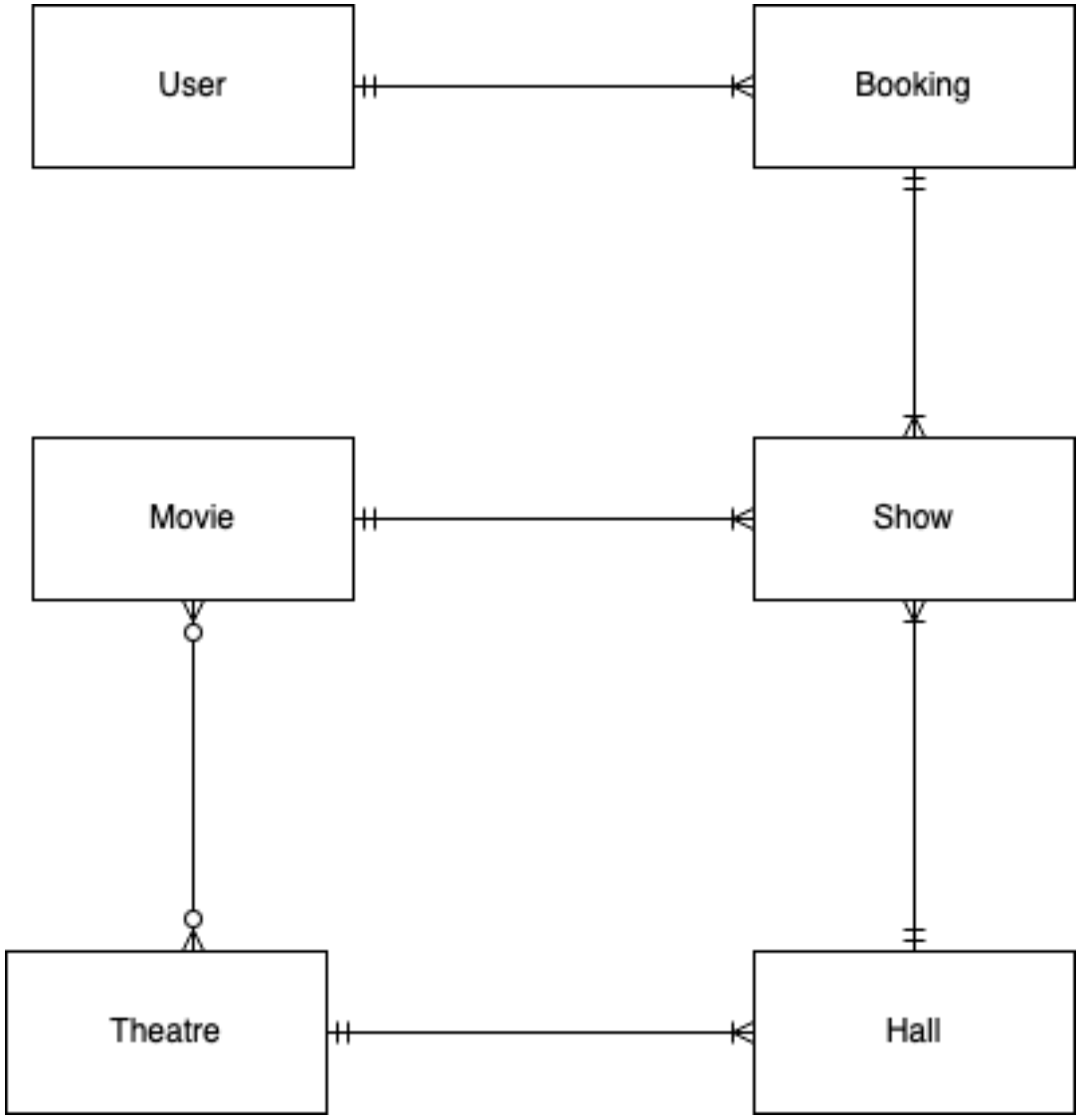

Figure 9 Initial ERD